

Lesson plan

Name of the Programme:	B.Tech(Hons.)	
Title of the Course:	Computer Networks	
Course code	CS2403	
Semester:	III	
Names of the Course Instructor(s)	Sheba Pari N., Sonam V. Maju, Neha Najem, Dr. Sumathi H.R., Lokesh J.K., Aishwarya Singh, Sunil Kumar J.	
Course credit	No. of hours per week (2+0+2)	Total no. of Teaching hours
03	04 Hours	60 Hours

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1.	1	Course syllabus overview, lesson plan, Brief history of Internet and its evolution	Lecture + Timeline activity (students draw evolution chart)	Peterson & Davie (2021) (Ch. 1), Stallings (2013) (Ch.1)
2.		Difference between asynchronous and synchronous communication with RS-232 and I2C examples	Chalk-talk + comparison table + live demo or video showing RS-232 vs I2C communication	Peterson & Davie (2021) (Ch. 2), Stallings (2013) (Ch.8)
3.		Lab Introduction Cisco Packet Tracer	Laboratory - Download and install CISCO packet tracer - Download and install Wireshark. - Introduction to hub, switch, router, gateways in CISCO packet tracer. - Activity 1: Tracing Route to Websites Using tracert. - Activity 2: Understanding IP Configuration on Windows Using ipconfig.	External Resources
4.				
5.		Client-server and peer-to-peer models; Distance-	Interactive discussion + Role play activity: groups act out client/server and P2P setups.	Peterson & Davie (2021) (Ch. 1, 2), Stallings (2013)

		based classification of networks, such as PAN, LAN, MAN, WAN, SAN	Diagram-based learning + Match-the-network quiz + classroom examples	(Ch. 2)
6.		Introduction to Internet access infrastructure in India. Introduction to the mobile communication technology evolution from 1G to 5G	PPT with coverage map evolution; Group discussion on mobile usage trends	Peterson & Davie (2021) (Ch. 2), Stallings (2013) (Ch. 8)
7.		Lab 1: Construct networks using switches and routers with hosts and test them using ping	Laboratory: Constructing the network in CISCO packet tracer and tracing the packets in Wireshark.	External Resources
8.				
9.		Node devices, connecting devices	Device identification activity (images or real hardware)	Peterson & Davie (2021) (Ch. 2), Stallings (2013) (Ch. 6)
10.		Network Devices (Router, Switch, Bridge, Hub, Gateways)	Use of network simulator tools	Peterson & Davie (2021) (Ch. 2), Stallings (2013) (Ch. 6)
11.		Lab 2: Connect three networks using a router with three interfaces, observe ICMP packets by connecting sniffers to a network with switches or routers	Laboratory: Constructing the network in CISCO packet tracer and testing the ping. View the packet ICMP in sniffers.	External Resources
12.				
13.	2	System requirements of the Internet; network performance parameters, such	Graphical explanation using tools like Wireshark; latency demo using online tools	Peterson & Davie (2021) (Ch. 3), Stallings (2013) (Ch. 2, 3)

		as bandwidth, latency, delay, jitter		
14.		Network byte order, generic network messages and headers format	Lecture + Discussion on real-world applications (VoIP, video conferencing) Dissect a network packet using Wireshark, group analysis of header fields	Peterson & Davie (2021) (Ch. 3), Stallings (2013) (Ch. 2, 4)
15.		Lab 3: Construct network using Ethernet hubs, bridges and observe packets using sniffers	Laboratory: Constructing the network in CISCO packet tracer and viewing packet format in the sniffers.	External Resources
16.				
17.		Layered architecture	Diagram walkthrough; student activity: design a 3-layer mini-model with annotations	Peterson & Davie (2021) (Ch. 1, 2), Stallings (2013) (Ch. 2)
18.		Interfaces between the layers and services provided	Interactive quiz per layer	Peterson & Davie (2021) (Ch. 1, 2), Stallings (2013) (Ch. 2)
19.		Lab 4: Understand ARP protocol using network constructed with hosts, switches and sniffers	Laboratory: Tracing the ARP packets through sniffers	External Resources
20.				
21.		Introduction to 7 layers of OSI model	Group roleplay: simulate data flow across OSI layers	Peterson & Davie (2021) (Ch. 1), Stallings (2013) (Ch. 2)
22.		TCP/IP Protocol suite	Video explanation + Comparative table creation + MCQ/Group reflection	Peterson & Davie (2021) (Ch. 1, 3), Stallings (2013) (Ch. 5)
23.		Lab 5: Understand IP header fields using networks with routers, switches, hosts and sniffers	Laboratory: View the IP header through sniffer and examine the source and destination addresses when ping from different pairs of source and destination.	External Resources
24.	3			
25.		Ethernet standard 802.3, CSMA/CD,	Lecture with animations of CSMA/CD; Draw and explain Ethernet frame MAC address	Peterson & Davie (2021) (Ch. 5), Stallings (2013)

		Ethernet frame structure	decoding game	(Ch. 14)
26.		MAC address format	MAC address numericals decoding	Peterson & Davie (2021) (Ch. 2), Stallings (2013) (Ch. 15)
27.		Lab 6: Understand and construct a network with DHCP servers and five routers using DHCP protocol	Laboratory: Examine through DHCP how IP addresses are being assigned to every device.	External Resources
28.				
29.		Wireless communications 802.11 frame structure, Wi-Fi, ISM, Bluetooth and piconet	Wi-Fi packet capture demo using Wireshark, ISM band spectrum chart, Explanation + case study: Bluetooth in smart homes, Group activity: form a mock piconet	Peterson & Davie (2021) (Ch. 5), Stallings (2013) (Ch. 15)
30.		Three types of switching technologies, datagrams, virtual circuits, source routing	Diagram-based explanation; comparison table; real-world examples (VoIP vs streaming)	Peterson & Davie (2021) (Ch. 6), Stallings (2013) (Ch. 12)
31.		Lab 7: Construct a network with five routers and configure the routers with default routes and RIP	Laboratory: After constructing the network, configure the distance between nodes through RIP	External Resources
32.				
33.	4	Internetwork, Internet Protocol (IP), IP service model, IP datagram delivery model; IPv4 header format	Diagram explanation of internetworking; real-life analogy activity (postal system)	Peterson & Davie (2021) (Ch. 3, 4), Stallings (2013) (Ch. 5)
34.		Need for fragmentation and reassembly and its implementation	Annotated breakdown of IP header fields; live packet capture demo showing fragmentation	Peterson & Davie (2021) (Ch. 4), Stallings (2013) (Ch. 5)
35.				

36.	Lab 8: Understand VLAN by constructing two VLANs on a single switch	Laboratory: Create 2 VLAN's and ping between 2 different VLANs and also between the same VLAN and compare the results.	External Resources
37.	Introduction and motivation for IPv6. IPv4 addressing schemes	Video comparison of IPv4 vs IPv6; class discussion: Why is IPv6 needed?	Peterson & Davie (2021) (Ch. 4), Stallings (2013) (Ch. 18)
38.	Classful addressing schemes, class A, B, C, D and E	IP class identification activity, worksheet with example addresses	Peterson & Davie (2021) (Appendix A, Ch. 4), Stallings (2013) (Ch. 19)
39.	Introduction to Networks Hardware Lab	Laboratory: Introduction to all the switches, management ports, uplinks, console ports, cables and connectivities between switch and hosts.	External Resources
40.			
41.			
42.			
43.			
44.	Lab 9: Configure an L2 Switch and understand its working using the Hardware lab	Laboratory: Boot switch as primary device (L2 switch) and then connect the host to the switches and launch puTTY app to work on the switch.	External Resources
45.	Address Resolution Protocol (ARP); Dynamic Host Configuration Protocol (DHCP), Internet Control Message Protocol (ICMP)	Wireshark demo showing ARP/DHCP traffic; flowchart creation by students, Hands-on: Use ping, tracert; case study on unreachable hosts	Peterson & Davie (2021) (Ch. 5), Stallings (2013) (Ch. 20)
46.	Network administration tools, ping,	Command-line hands-on session + Group lab work,	Peterson & Davie (2021) (Ch. 5),

		tracert. Motivation behind subnetting and classless addressing	troubleshooting a mock network	Stallings (2013) (Ch. 5)
47.	5	Lab 10: Understand Link aggregation using the Hardware lab	Laboratory: Connecting data ports according to the link aggregation required.	External Resources
48.				
49.		UDP Protocol, UDP header Transmission Control Protocol (TCP)	Header field breakdown using diagrams, analyze a UDP packet in Wireshark 3-way handshake animation	Peterson & Davie (2021) (Ch. 6), Stallings (2013) (Ch. 10)
50.				
51.		Lab 11: Router on a Stick configuration using the Hardware lab	Laboratory: Configure the switch as secondary memory to boot as a router and configure the router	External Resources
52.				
53.		TCP header	Analyze a TCP packet in Wireshark	Peterson & Davie (2021) (Ch. 6), Stallings (2013) (Ch. 10)
54.		TCP Vs UDP	TCP vs UDP comparison chart activity	Peterson & Davie (2021) (Ch. 6), Stallings (2013) (Ch. 10)
55.		Lab 12: Inter- VLAN routing using the Hardware lab	Laboratory: Create VLANs and try communicating between 2 different VLANs and also within the same VLAN, with and without using uplinks.	External Resources
56.				
57.		Routing Information Protocol (RIP)	RIP routing table walkthrough; route update simulation (on paper or simulator)	Peterson & Davie (2021) (Ch. 7), Stallings (2013) (Ch. 12)
58.		VLAN, VPN, NAT and VoIP	Packet Tracer demo for VLANs, IP address translation activity, NAT configuration examples, Case study: VoIP apps (Skype, WhatsApp); simulate latency/jitter impact on voice quality	Peterson & Davie (2021) (Ch. 7), Stallings (2013) (Ch. 18, 22, 24)
59.		MINI PROJECT REVIEW	Laboratory- Group-wise review the project 1 st review	External Resources
60.				

Assessment and Evaluation

Continuous Internal Evaluation (CIE): 70%	
Components of CIE	Weightage of each component
CIE-1 - Quiz Graded Component 1 - - IBM SkillsBuild Courses "Computer Networking Basics: How Computers Communicate", https://skills.yourlearning.ibm.com/activity/MDL-236 Graded Component 2	15 marks out of 70 5 marks out of 70
CIE-2 - Theory	25 marks out of 70
CIE-3 - Lab execution, verification of output & viva-Component 1 - Final Project Submission- Component 2	10 marks out of 70 15 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Theory
Weightage of Exam	30%

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Component 1	Quiz will be conducted for 15 marks. Each right answer carries one mark. No negative marking for wrong answers.
	Component-2	Certificate of completion for 5 marks. The specified courses on the IBM SkillsBuild platform are to be uploaded to the designated Google Classroom. (Refer Table-1)

Table-1:

Graded Component-2 (5 Marks):

Assess students' ability to complete assigned courses on the IBM SkillsBuild platform and reflect on their learning.

Criteria	Excellent (5 Marks)	Satisfactory (3 Marks)	Needs Improvement (0 Marks)
Completion of Courses (5 Marks)	Successfully completes all specified courses on the IBM SkillsBuild platform before the deadline. Certificates are uploaded to Google Classroom with no errors.	Completes some courses or has minor errors in uploading certificates.	Fails to complete the courses or does not upload certificates.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Component 1	Continuous Lab assessment + Viva = 10 marks <i>(Refer Table-2)</i>
	Component 2	Course Project + Report = 15 marks <i>(Refer Table-3)</i>

Table-2

Lab Write-up and Execution rubrics (Max: 6 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0 Mark)
1.	Understanding of problem statement. (2 Marks)	Observations	Student exhibits thorough understanding of requirements and applies suitable algorithm for the problem (2 M)	Student has sufficient understanding of requirements and applies suitable algorithm for the problem. (<2 M and >=1 M)	Student does not have a clear understanding of requirements and is unable to apply suitable algorithm for the problem. (0 M)
2.	Execution (2 Marks)	Observations	Student demonstrates the execution of the program with optimized code and shows performance efficiency. Appropriate validations with all test cases are handled. (2 M)	Student demonstrates the execution of the program without optimization of the code and shows performance efficiency with only few test cases. (1 M)	Student has not executed the program. (0 M)
3.	Results and Documentation (2 Marks)	Observations	Documentation with appropriate comments and output is covered in data sheets and manual. (2 M)	Documentation with only few comments and only few output cases is covered in data sheets and manual. (1 M)	Documentation with no comments and no output cases is covered in data sheets and manual. (0 M)

Viva Voce Rubrics (Max: 4 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0 Mark)
1.	Conceptual Understanding (2 Marks)	Viva Voce	Explains thoroughly the network design and the components used along with related concepts. (2 M)	Adequately explains the network design and the components used with related concepts (1 M)	Unable to explain the network design and the components used (0 M)
2.	Communication of networking Concepts (2 Marks)	Viva Voce	Communicates the concept used in problem solving well. (2 M)	Sufficiently communicates the concepts used in problem solving. (1 M)	Unable to communicate the concepts used in problem. (0 M)

Table-4

Grading Rubrics for CN Group Projects (Total: 15 Marks)					
Sl. No	Criteria	Excellent	Very Good	Good	Needs Improvement
1.	Report Submission (5 Marks)	Comprehensive, well-organized, and insightful report that covers all aspects in depth. (5 M)	Thorough report with minor areas for improvement. (3-4 M)	Report covers all essentials but lacks detail or organization. (2 M)	Incomplete or significantly flawed report. (0-1 M)
2.	Project Demonstration (4 Marks)	Flawless execution demonstrating Networking problems (Hardware or software) techniques. (4 M)	Effective implementation with minor technical issues. (3 M)	Functional implementation with some errors or omissions. (2 M)	Lack of demonstration or significant errors present. (0 M)

3.	Peer Review (3 Marks)	Exceptional engagement providing and incorporating constructive feedback. (3 M)	Adequate feedback that adds value, with slight omissions. (2 M)	Basic feedback provided, lacks depth or critical analysis. (1 M)	No feedback provided or feedback is not constructive. (0 M)
4.	Presentation and Viva (3 Marks)	Outstanding delivery and depth of knowledge displayed in Q&A. (3 M)	Solid presentation and good comprehension shown in Q&A. (2 M)	Adequate presentation with some misunderstandings in Q&A. (1 M)	Presentation is incomplete or fails to address Q&A adequately. (0 M)